Shrishti Tomar

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Summary

Software Developer and Computer Science undergraduate (B.Tech, 2027) focused on building scalable full-stack and backend systems with Node.js, Express, PostgreSQL, Redis, and Docker. Deployed three production projects on AWS EC2, including an asynchronous job queue engine and a RAG-based code search platform.

Education

Guru Gobind Singh Indraprastha University — Delhi, India
B.Tech, Computer Science and Engineering; CGPA: 8.29 / 10 — *Expected June 2027*
Class XII; Percentage: 94.6%

Projects

RepoInsight | AI Code Search Platform — GitHub | Live

- Built a **Retrieval-Augmented Generation (RAG)** platform that clones public GitHub repositories and indexes source files as **vector embeddings**, enabling natural-language search across codebases of up to **10,000+ files**.
- Designed a **LangGraph** multi-agent layer routing queries across code search, dependency mapping, and summarization agents, cutting average answer latency from **12s to 3s**.
- Containerized with a multi-stage **Docker** build packaging the Vite frontend and Express backend into a single container **(size down 85%)** and deployed to **AWS EC2**, with **JWT** auth and chat history persisted in **MongoDB Atlas**.

TaskForge | Distributed Task Queue Engine — GitHub | Live

- Engineered an asynchronous task processing engine using **Express, BullMQ, and Redis** to offload CPU-intensive operations (OCR extraction, PDF generation, scraping) across **5 concurrent worker threads** with **sub-1.8ms** Redis transport latency.
- Implemented a **Socket.IO** WebSocket telemetry engine broadcasting live job progress (0% to 100%) and audit logs to client dashboards, replacing HTTP polling with server-pushed updates.
- Orchestrated a **5-container microservices** stack via **Docker Compose**, modeled **PostgreSQL** schemas using **Prisma ORM** with exponential backoff retries (max 3 attempts), and implemented dual-token **JWT** authentication with **RBAC**.

DropAlert | E-Commerce Price Tracker — GitHub | Live

- Built a price tracker that extracts structured product data from e-commerce listings via the **Firecrawl API**, bypassing anti-bot protections that block conventional scrapers.
- Automated periodic price checks with **cron jobs** and **Redis** caching across **50+ tracked products**, comparing results against user-defined targets.
- Configured templated transactional email alerts via **Resend**, notifying users within **5 minutes** of a price drop, and charted price history.

Technical Skills

Languages: C++, Java, JavaScript, TypeScript, Python, SQL
Frontend: React, Next.js (App Router, SSR), Redux Toolkit, TanStack Query, Tailwind CSS
Backend: Node.js, Express.js, REST APIs, BullMQ, WebSockets (Socket.IO), JWT Authentication
Databases: PostgreSQL, MongoDB, Redis, Supabase, Prisma ORM
AI: Retrieval-Augmented Generation (RAG), Vector Embeddings, Gemini API
Cloud and DevOps: AWS EC2, Docker, Docker Compose, Git/GitHub, Linux, CI/CD Pipelines

Achievements

- **GTBIT Hackathon:** Secured 3rd place among 80+ competing teams.
- **HackMait Hackathon:** Placed in the top 17 teams out of 700+ participants.
- **Competitive Programming:** Solved 400+ Data Structures and Algorithms problems on LeetCode and other platforms.
- **Namespace Community:** Junior Counselling Member.